

Class Exercise 5: Experimental Design Planning

Instructions: **Brainstorm** ideas for our class experiment: replication and extension of Ludwig et al., (2022).

Requirements: To successfully complete this exercise, you are required to:

1. Work with your partner to **reflect** on two IH manipulations
2. **brainstorm** our IV #2 manipulation for our framing experiment

Please use a **word processing program** (such as MS Word) to complete this exercise.

- - - - -

IH Manipulations

Fallibility Salience Manipulation: Using this method, participants are randomly assigned to an IH condition or a control condition.

In the IH condition, participants answer three science-related multiple-choice questions with seemingly obvious answers: “What weighs more, a pound of gold or a pound of feathers?” “What mountain peak is farthest from the center of the Earth?” and “How many senses do humans have? Participants then write and reflect on why it is okay to recognize that their knowledge and beliefs may be inaccurate.

In the control participants also complete three multiple-choice questions. However, all questions in the control condition are about office technology (e.g., “What office technology do you use most for work?”). Participants in the control condition also write and reflect on why it is okay to use office technology to make certain aspects of work more efficient.

When you completed the IH exercise, how did it make you feel, and how do you think it will make participants feel?

[After seeing that I was incorrect for all three questions, I kind of laughed at myself. I trusted that the answers given were correct but was interested in why they were correct. I’m not sure if other participants would also laugh at themselves, but I am sure that this exercise will get them

to think about why they could be wrong.]

Is it possible that this manipulation could create a negative experience for some participants? Could this manipulation even backfire and not have the intended temporary boost in intellectual humility? Why?

[For me personally, I was dissatisfied with the answer for the first question. I understood that in technical terms, a pound of feathers weighs more. However, I have seen this question presented before, and the correct answer was that they weigh the same. So, these two answers are contradicting one another in my head.

However, I think that this is just a flaw with the question. In general, I think that when most people are presented with questions that do not affect them personally, they are willing to accept that they could be incorrect and engage with IH]

Reflection Exercise: Alternatively, many of you suggested that we can manipulate (i.e., operationalize) intellectual humility by having participants complete an IH reflection exercise (vs. a control condition).

You will now complete a rough draft of the IH reflection task. First, complete the survey at the link below, and then answer the questions that follow:

https://pitt.co1.qualtrics.com/jfe/form/SV_2nKzZZB3vuodXBs

Note: In the actual study, participants will first be randomly assigned to either the IH or control condition. For this example, you will see both.

After the study information page, participants in the IH condition complete the reflection exercise.

How easy or difficult was it for you to think of a time when you acted with intellectual humility?

[It took me a few minutes to think, but I was able to think of a time pretty quickly.]

The IH prompt included the following sentence:

“Do not describe an event in which you felt humiliated or shameful; rather, describe when you acted with intellectual humility or a situation where you had a mindset consistent with the characteristics of intellectual humility described above.”

Why do you think it might be important to include this instruction?

[Because humiliation and shame can steer somebody to act out of self-defense. Sometimes, the easiest thing to do when those feelings are invoked is to avoid conflict. This is important because the question aims to find a time where the participant naturally acted out of IH]

Do you think the fallibility salience or reflection exercise is a better manipulation of IH?

[The reflection exercise]

Can you think of any improvements to this prompt? Or would this exercise be effective at temporarily increasing a state of intellectual humility?

[A potential improvement could be letting the participants know which answers demonstrate intellectual humility once they complete the survey.]

For our class experiment, we want to *replicate* Ludwig and colleagues (2022) using an experimental design. We will therefore experimentally manipulate intellectual humility by randomly assigning participants to either an IH or a control condition. However, we will *extend* Ludwig and colleagues' results by focusing on a *new* second IV. As in the original study, apology behavior will be our DV.

Brainstorm IV #2

Now, your task is to think of other variables that could also influence whether someone apologizes and whether this variable might interact with intellectual humility

In Ludwig et al. (2022), the researchers measured **intellectual humility (IH)** and manipulated the **severity of the offense**. Participants were randomly assigned to imagine committing either a **high-severity** or **low-severity** offense. The goal was to examine whether IH is especially important for predicting apologies in certain situations.

Besides severity, what other variable(s) might affect whether someone is willing to apologize? Why?

[The number of people that are present for the apology]

Brainstorm two variables and describe how you would **manipulate** them in an experiment. Importantly, because of our study design, this variable needs to have two levels.

[Political affiliation (republican or democrat)]

Gender of the person receiving apology (Male or Female)]

Below are some examples:

- Public vs. private setting (apologizing in front of others vs. one-on-one)
- Intentional vs. accidental harm
- Power/status difference (apologizing to a boss vs. a subordinate)
- Relationship closeness (high closeness vs. low closeness)
- Victim's initial response (More forgiving vs. less forgiving)

